

Use the equation $y = -\frac{1}{2}x - 3$ for questions 1 and 2.

1. What is the form of this linear equation?

This equation is in slope-intercept form.

2. Rewrite this equation in standard form.

$$y = -\frac{1}{2}x - 3$$

$$+\frac{1}{2}x \quad +\frac{1}{2}x$$

$\frac{1}{2}x + y = -3$ or any
equivalent equation in
standard form.

Use the equation $y - 7 = \frac{2}{3}(x - 9)$ for questions 3 and 4.

3. What is the form of this linear equation?

This equation is in point-slope form.

4. Rewrite this equation in standard form.

$$\begin{aligned} y - 7 &= \frac{2}{3}(x - 9) \\ y - 7 &= \frac{2}{3}x - \frac{2}{3} \cdot \frac{9}{1} \\ y - 7 &= \frac{2}{3}x - 6 \end{aligned}$$

$-\frac{2}{3}x + y = 1$ or any
equivalent equation in
standard form.

Use the equation $3x + 5y = -2$ for questions 5 and 6.

5. What is the form of this linear equation?

This equation is in standard form.

6. Rewrite this equation in slope-intercept form.

$$\begin{array}{rcl} 3x + 5y = -2 & \frac{5y}{5} = \frac{-3x}{5} - \frac{2}{5} \\ \underline{-3x} & \underline{-3x} & \end{array}$$

$$5y = -3x - 2 \quad \boxed{y = \frac{-3}{5}x - \frac{2}{5}}$$

Use the equation $y + 1 = 2(x + 4)$ for questions 7 through 10.

$$y - y_1 = m(x - x_1)$$

7. What is the slope of the line?

$$\begin{array}{rcl} y + 1 = 2x + 8 \\ y = 2x + 7 \\ y = mx + b \\ m = 2 \quad b = 7 \end{array}$$

The slope is 2.

8. Rewrite this equation in slope-intercept form.

$$\boxed{y = 2x + 7}$$

9. Identify the y -intercept of the line.

The y -intercept is (0,7)

10. Rewrite the equation in standard form.

$$\begin{array}{rcl} y + 1 = 2x + 8 & y = 2x + 7 \\ \underline{-1} & \underline{-1} & \underline{-2x} \quad \underline{-2x} \end{array}$$

$-2x + y = 7$ or any equivalent equation in standard form.